
How Much of the Elliptic Leaderboard Is Real? Decomposing Evaluation Inflation into Leakage, Shift, and Base-Rate Effects on Temporal Transaction Graphs

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Abstract

The Elliptic Bitcoin dataset and its extensions are standard benchmarks for graph-based fraud detection, and recent work reports illicit-detection scores approaching 0.98. We show that these numbers are largely an artifact of the evaluation protocol. Holding models, features, and training fixed, a random split inflates illicit-class PR-AUC by +0.195 (XGBoost) to +0.553 (GCN) over five paired seeds, relative to the standard temporal split. Moreover, the full published recipe, a random split combined with aggregate metrics, reproduces the reported headline numbers with a generic XGBoost (accuracy 0.992, weighted F1 0.992). We then decompose this gap with two controls. First, a prevalence-matched random test set isolates a small base-rate component. Second, an inductive ablation, which removes all test nodes from the graph during training, shows that the message-passing leakage we initially hypothesized is indistinguishable from zero for all four GNN architectures tested, including a graph transformer. The dominant component is *distribution access*: a random split trains the model on the test period’s distribution, and the models that generalize worst across temporal shift gain the most. Consistent with this mechanism, the inflation tracks the drift magnitude across four datasets (stable +0.001, moderate drift +0.124, regime change +0.41 to +0.55) and concentrates in the post-shift window. Under the temporal protocol, gradient-boosted trees still lead graph neural networks, and four drift-robust variants, including a first adaptation of USAD to graphs, fail to improve on the baselines. Every reported number can be regenerated with a single seeded command from our repository.

1 Introduction

Detecting illicit activity in blockchain transaction graphs is a canonical application of graph machine learning. The Elliptic dataset [1] (203,769 Bitcoin transactions across 49 time steps, ~2% illicit) and Elliptic++ [2] (adding 822,942 wallet addresses) are the de facto benchmarks. A stream of recent papers reports illicit-detection scores approaching 0.98, for example accuracy 0.9802 and weighted F1 0.9799 under a stratified random split [3], or random-split evaluations on Elliptic++ [4]. This suggests the problem is nearly solved. We argue this impression is false and traces to two evaluation choices: (i) random rather than temporal train/test splits, which give models access to the test period’s distribution, and (ii) reporting weighted or overall F1, dominated by the ~98% licit class, rather than illicit-class metrics.

This question matters to the graph learning community specifically. Elliptic is one of the most cited node-classification benchmarks with real-world stakes, and the gap we quantify is largest precisely for GNNs: the models this community builds gain the most from the broken protocol. An evaluation regime that systematically favors a model family distorts the apparent state of the art and the direction of methods research.

Our contributions are: (1) a systematic, multi-model, multi-seed quantification of the random-split inflation on Elliptic-family benchmarks; (2) a decomposition of that inflation into base-rate, graph-leakage, and distribution-access components, including the measured refutation of our own message-passing-leakage hypothesis; (3) a temporally stable cross-domain control consistent with the shift-driven mechanism; and (4) negative results, reported in full, for four drift-robust variants. These negatives indicate that the post-shift regime is the genuinely open problem, rather than an unexplored one. We release an evaluation harness in which every reported number is one seeded command.

2 Related work

Evaluation integrity. Leakage was formalized by Kaufman et al. [5] (the “no time machine” rule); Kapoor and Narayanan [6] taxonomize it as a driver of ML reproducibility failures; Arp et al. [7] name “temporal snooping” among security-ML pitfalls; Shchur et al. [8] document protocol sensitivity in GNN evaluation generally. The closest template is TESSERACT [9], which showed Android-malware F1 of ~ 0.99 collapsing under time-aware evaluation. Concurrently, Roth [10] quantifies four leakage classes across 2,047 *iid* tabular benchmarks, finding normalization leakage negligible and boundary effects invisible under random cross-validation. This is precisely the regime where our mechanism predicts no inflation. Our results show the complementary regime: on temporally drifting fraud benchmarks the split protocol becomes the dominant effect, through distribution access rather than any classical leakage class. Relative to this line, our contribution is graph-native and mechanistic: we quantify the inflation on Elliptic-family benchmarks across model classes and decompose it, showing the graph-specific channel is null and the dominant term is distribution access.

Temporal graph benchmarks. The Temporal Graph Benchmark [11] standardizes chronological splits for temporal graph learning tasks, and Chartalist [12] provides labeled blockchain graph datasets with timestamps that make time-aware splits possible. Our results supply a quantified argument for why this discipline is not optional on fraud graphs: we measure what is gained, artificially, when it is dropped, and show that the gain is largest for the models whose temporal generalization is weakest.

Elliptic-specific evaluation. Weber et al. [1] established the temporal protocol and the trees-over-GNNs result; GADBench [13] confirms tuned tree ensembles lead on Elliptic (tuned RF AUPRC ≈ 0.789 , matching our XGBoost). Closest to our ablation, a recent preprint [14] reports a large (~ 39.5 F1 point) transductive-versus-inductive protocol gap on Elliptic under temporal evaluation: training-time exposure to test-period graph structure inflates results. This is consistent with our decomposition, which offers a candidate explanation rather than standing in tension with our null: under a *temporal* protocol, test-period nodes are the model’s only access to the shifted test distribution, so graph visibility carries exactly the distributional information that drives inflation; under our *random-split* ablation, the labeled training set already spans all periods, test-node features are redundant, and removing them changes nothing (-0.000 ± 0.008). On this reading the graph channel is not a separate leak: it matters precisely insofar as it transports otherwise-unavailable distributional information. In other words, a single mechanism expresses itself under two protocols. A direct test (our inductive ablation run under the temporal protocol) would adjudicate this reading and is left to future work.

Concurrent work: construction-level leakage. While this study was underway, Šafář et al. [15] published a forensic deconstruction of the Elliptic dataset itself: reverse-engineering over 90% of its feature semantics, they show that the “local” features embed address-level global statistics computed at a reference time *after* the test period (look-ahead by construction), that feature normalization was fit on the full dataset, and that shared addresses couple training and test samples through both features and inherited labels. Their axis is complementary to ours: they identify and formalize *construction-level* leakage without quantifying its performance impact; we quantify and decompose *protocol-level* inflation without touching feature construction. The two compose: both arms share the same features, so *exposure* to construction-level leakage is symmetric, though its induced *effect* need not be identical across arms; we therefore treat it as an unmeasured layer that makes our temporal-arm scores *upper bounds* on deployable performance and could perturb individual components (flagged in Limitations), while leaving the qualitative decomposition intact. The true deployment-realistic numbers are therefore lower still, which strengthens rather than weakens the overall conclusion. None of the works above measure random-split inflation across models, decompose it into components, or test drift-graded controls.

Table 1: Datasets. Drift regime is characterized in Section 4.5: Elliptic’s labeled fraud prevalence collapses from 9.2% to 2.5% at the dark-market shutdown (time step 43); AMLworld’s laundering prevalence doubles across the stream; DGraph-Fin’s fraud rate is temporally stable.

Dataset	Scale	Task	Split	Drift regime
Elliptic [1]	204k tx, 234k edges	tx classification	time step 34	regime change
Elliptic++ [2]	+823k addresses, 2.87M edges	address classification	first-seen step	regime change
DGraph-Fin [18]	3.7M nodes	node classification	first-seen time	stable
IBM AMLworld HI-Small	5.08M tx (synthetic)	tx classification	transaction time	moderate

Graph anomaly detection and USAD.. Reconstruction autoencoders [16], generative and adversarially regularized variants, and one-class methods form the unsupervised graph-anomaly literature. Among adversarial reconstruction schemes, we selected USAD [17], a two-decoder autoencoder for multivariate time series, as the representative approach for our negative-result study: it is explicitly designed for training stability over GAN-style alternatives, which makes it a strong candidate under a regime where training instability could otherwise explain a failure. To our knowledge it had not previously been adapted to graphs; we adapt it and report the outcome.

3 Experimental protocol

All experiments use one codebase, fixed seeds, and a CI-gated environment (Table 1). The evaluation protocol is: a temporal split (Elliptic’s canonical early-versus-late time steps; first-seen splits for addresses and DGraph-Fin), illicit-class PR-AUC and F1 as primary metrics, a validation slice carved from the latest train steps, and decision thresholds chosen on validation only. Leakage comparisons hold the model, features, and trainer fixed and change only the split. Models: GCN [19], GraphSAGE [20], GAT [21], a graph transformer (two TransformerConv layers, the UniMP operator [22], 4 heads), and XGBoost [23]. Unless stated otherwise, GNNs use two convolution layers, hidden dimension 64, dropout 0.5, Adam with learning rate 10^{-2} and weight decay 5×10^{-4} , up to 200 epochs with early stopping on validation PR-AUC (patience 20), and a class-weighted cross-entropy loss; XGBoost uses 300 trees of depth 6 with `scale_pos_weight` set from the train class ratio. Features are standardized with a scaler fit on the training arm only.

Statistics. Multi-seed results use five paired seeds (42–46) unless stated otherwise; we report mean $\pm$ sample standard deviation (ddof= 1) and, for headline claims, 95% t -confidence intervals and exact sign-flip permutation tests on the paired per-seed differences (the smallest attainable one-sided p at $n=5$ is $1/32$). The “leakage is null” statement is an equivalence claim and is therefore assessed with two-one-sided tests (TOST): we report the smallest round symmetric margin containing the 90% CI; these margins are descriptive (post hoc), not pre-registered. Per-seed values for every cell are released with the code (Appendix A). The Elliptic++ address task, the DGraph-Fin control, DANN, and the USAD variants use three paired seeds; the per-architecture tuning sweeps, the EvolveGCN sweep, the DGraph width-128 probe, and the step-2 full-train baseline are single runs at seed 42 and are labeled as such where quoted.

4 Results

4.1 Baseline ordering under the temporal protocol

Under the temporal split, XGBoost (PR-AUC 0.791 ± 0.002 ; F1 0.82 at seed 42 under the full-train protocol) leads GraphSAGE (0.510 ± 0.039), the graph transformer (0.487 ± 0.038), GAT (0.297 ± 0.061) and GCN (0.265 ± 0.058), reproducing Weber et al. [1]. Transformer-style dot-product attention does not close the temporal generalization gap: it lands within seed noise of GraphSAGE, far below the tree baseline. A 12-configuration tuning sweep per architecture, selected on temporal validation only (seed 42), does not change this picture: for GraphSAGE it improves F1 ($0.422 \rightarrow 0.486$) while leaving PR-AUC within seed noise of the default (0.506 vs 0.510 ± 0.039); for GCN, validation selection actually *degrades* test PR-AUC ($0.294 \rightarrow 0.274$), an instance of validation-test misalignment under shift that we report rather than omit. Our GNN F1 remains below some honest-protocol literature values (~ 0.69); we report this openly as a protocol/training-detail discrepancy rather than claiming parity.

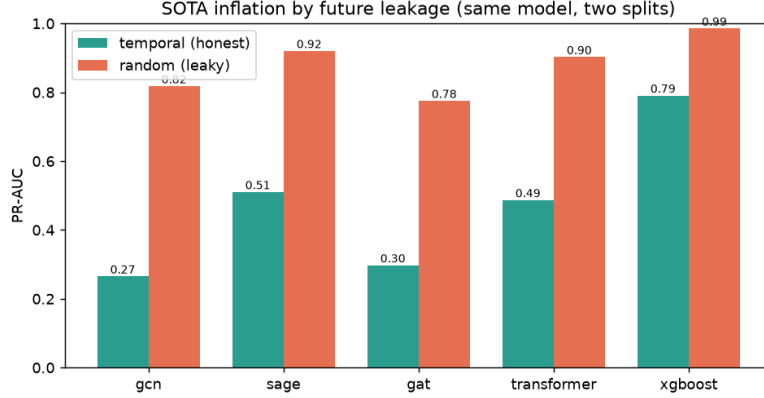


Figure 1: Temporal versus random-split PR-AUC per model (5 seeds). The inflation is largest for the GNNs, the models whose temporal generalization is weakest.

4.2 The inflation, quantified

Table 2: Illicit-class PR-AUC under the temporal split versus a stratified random split (mean $\pm$ std over 5 seeds). Same model, same trainer, same features; only the split differs.

Model	Temporal	Random	Inflation
GCN	0.265 ± 0.058	0.818 ± 0.010	$+0.553 \pm 0.062$
GraphSAGE	0.510 ± 0.039	0.920 ± 0.008	$+0.410 \pm 0.046$
GAT	0.297 ± 0.061	0.775 ± 0.036	$+0.478 \pm 0.041$
GraphTransformer	0.487 ± 0.038	0.902 ± 0.007	$+0.415 \pm 0.039$
XGBoost	0.791 ± 0.002	0.986 ± 0.001	$+0.195 \pm 0.001$

Every inflation is significant at the smallest attainable exact permutation level ($p = 1/32$), with 95% CIs bounded away from zero (e.g. GraphSAGE [0.353, 0.467]; Figure 1). Under the random split, XGBoost reaches PR-AUC 0.986 ± 0.001 (F1 0.955 at seed 42), inside the literature’s reported ~ 0.90 – 0.98 band.

The published recipe, replicated exactly. Following the full published recipe, a stratified random split combined with aggregate metrics, a generic XGBoost reproduces the headline numbers (5 seeds): accuracy 0.9917 ± 0.0002 and weighted F1 0.9916 ± 0.0002 , matching and slightly exceeding reported values such as accuracy 0.9802 / weighted F1 0.9799 [3]. We replicate the protocol class (split and metric choices), not any specific pipeline’s implementation; this shows that the protocol alone suffices to reach the band. The same model under the temporal protocol scores illicit-F1 0.756 ± 0.007 / PR-AUC 0.791 ± 0.002 . Notably, *accuracy remains 0.969 even under the temporal split*: metric choice and split choice each contribute to the illusion, and together they turn 0.76 into 0.99 with no novel architecture involved. The effect generalizes to the Elliptic++ address task (heterogeneous graph, 822k address nodes) under a deployment-honest feature protocol (pre-split wallet aggregation, train-only scaling): over three seeds, PR-AUC 0.417 ± 0.052 (temporal) versus 0.965 ± 0.002 (random), an inflation of $+0.548 \pm 0.050$.

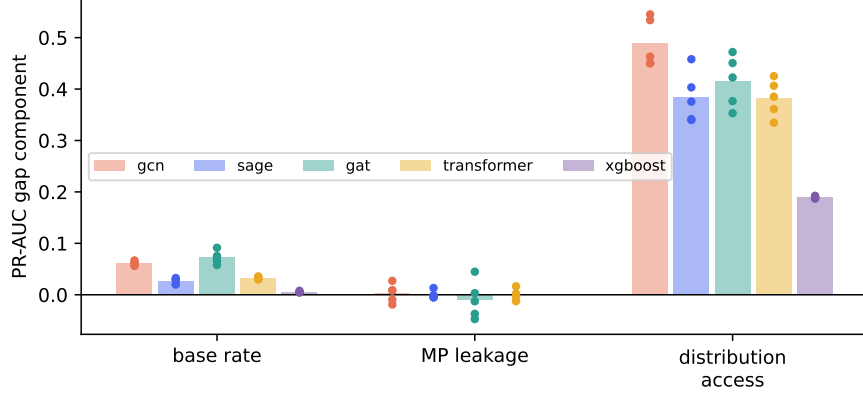
4.3 Decomposing the gap

Two controls, threshold-free PR-AUC, five seeds. (i) Matching the random test set’s prevalence to the temporal test’s (6.5%) isolates the base-rate component. (ii) An inductive ablation removes all test nodes from the graph during training, so no test feature or test-adjacent edge is visible; this measures graph-specific message-passing leakage directly.

The message-passing channel we ourselves hypothesized is **indistinguishable from zero for all four architectures**: training with test nodes entirely removed from the graph leaves random-split PR-AUC statistically unchanged. Because this is an equivalence claim, we assess it with TOST: the 90% CIs

Table 3: Decomposition of the temporal-versus-random gap (PR-AUC, 5 seeds).

Component	GCN	GraphSAGE	GAT	Transformer	XGBoost
Base rate	$+0.061 \pm 0.004$	$+0.026 \pm 0.006$	$+0.073 \pm 0.013$	$+0.033 \pm 0.002$	$+0.005 \pm 0.001$
Message-passing leakage	$+0.003 \pm 0.018$	-0.000 ± 0.008	-0.010 ± 0.036	$+0.000 \pm 0.011$	n/a
Distribution access	$+0.489 \pm 0.047$	$+0.384 \pm 0.049$	$+0.415 \pm 0.050$	$+0.383 \pm 0.036$	$+0.189 \pm 0.002$
Total gap	$+0.553 \pm 0.062$	$+0.410 \pm 0.046$	$+0.478 \pm 0.041$	$+0.415 \pm 0.039$	$+0.195 \pm 0.001$

**Figure 2:** Decomposition of the temporal-versus-random PR-AUC gap. Bars are 5-seed means; dots are individual seeds. The message-passing leakage component clusters at zero for all four GNNs; distribution access dominates everywhere.

bound the effect within ± 0.01 for GraphSAGE ($[-0.008, +0.007]$), ± 0.02 for the graph transformer ($[-0.010, +0.010]$), ± 0.03 for GCN ($[-0.014, +0.020]$), and ± 0.05 for GAT ($[-0.045, +0.025]$); the GAT bound is loose owing to its seed variance, so we can rule out a large leak but not a small one there. No architecture shows a positive leak. Distribution access is computed as the residual (total gap minus the base-rate and message-passing components), so its uncertainty compounds theirs. The dominant term everywhere is distribution access: a random split trains the model on examples drawn from the test period, an advantage no deployed model can have. GNNs inflate more than XGBoost (paired GraphSAGE–XGBoost inflation difference $+0.215$, 95% CI $[0.158, 0.272]$, $p = 1/32$) because their temporal generalization is weaker, not because of a graph-specific leak. We report this as an explicit self-correction. This null is also consistent with the large transductive-versus-inductive gap reported by Maganti [14] under temporal evaluation (Section 2), on the candidate reading that graph visibility inflates results only when it is the sole conduit of test-period distributional information; when distribution access is already granted by the split, the graph channel adds nothing measurable.

4.4 The mechanism’s falsifiable prediction: inflation concentrates in the shifted window

If distribution access drives the inflation, it should be largest where the distribution shift is largest: the post-shutdown window (steps ≥ 43), where the labeled fraud prevalence collapses from 9.2% to 2.5%.

Table 4: Within-window inflation (GraphSAGE, PR-AUC, 5 seeds). The random-over-temporal ratio cancels each window’s base rate.

Window	Temporal	Random	Inflation	Ratio
Pre-shutdown (35–42)	0.651 ± 0.042	0.895 ± 0.018	$+0.244 \pm 0.059$	$1.4\times$
Post-shutdown (≥ 43)	0.042 ± 0.001	0.348 ± 0.033	$+0.306 \pm 0.034$	$8.2\times$

The absolute difference is directional but not conclusive (paired $+0.062 \pm 0.058$, sign-flip $p = 0.0625$, 4/5 seeds). On the relative scale the separation is clear: the within-window random-over-temporal

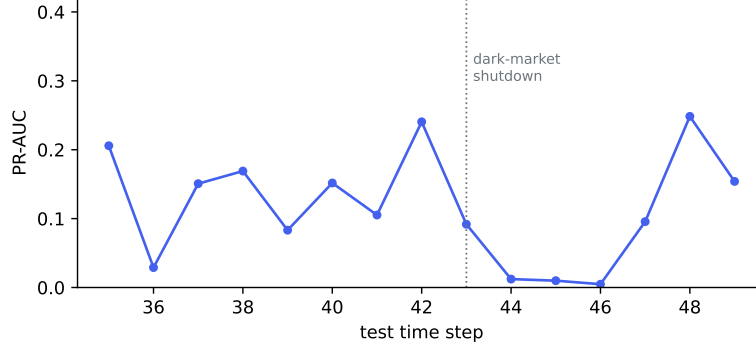


Figure 3: Rolling per-window test PR-AUC of EvolveGCN-O under the temporal protocol. Aggregate scores hide the collapse at the dark-market shutdown (steps 44–46).

ratio rises from $1.4\times$ pre-shutdown to $8.2\times$ post-shutdown, and the paired difference of per-seed log-ratios is $+1.78 \pm 0.10$ with all five seeds positive (sign-flip $p = 1/32$). Post-shutdown, the temporal model is close to the no-skill floor (0.042 vs a 0.025 base rate) while the random-split model retains substantial skill (0.348). This skill can only have been acquired by training on the post-shutdown period itself. Two caveats: the post-shutdown ratio’s denominator is near the floor, so the ratio’s magnitude (though not its sign) is sensitive to small absolute changes; and the two windows differ in base rate (9.2% vs 2.5%), so cross-window comparisons of *absolute* inflation are imperfect. The result is consistent with the mechanism’s prediction in the relative sense; the weaker absolute test is reported alongside.

4.5 Cross-domain controls: a dose-response pattern

Two additional datasets bracket the drift axis. On DGraph-Fin [18] (3.7M nodes, timestamped edges, temporally stable $\sim 1.3\%$ fraud rate), the same GraphSAGE shows essentially no PR-AUC inflation ($+0.001 \pm 0.001$ over three seeds): where the distribution does not shift, there is nothing to access. A wider model (hidden dimension 128) behaves identically (PR-AUC inflation $+0.004$; ROC-AUC $+0.055$). Caveats stated plainly: ROC-AUC does inflate ($+0.025 \pm 0.006$; up to $+0.055$ at width 128), both arms sit near the PR-AUC floor at this model strength, and GADBench reports strong GNNs on DGraph-Fin, so this control is supporting evidence rather than proof.

On IBM AMLworld HI-Small (5.08M synthetic transactions with complete ground truth, 0.102% laundering; tabular XGBoost, three seeds), whose generator drifts moderately (laundering prevalence $0.075\% \rightarrow 0.152\%$ across the stream), the inflation is intermediate: temporal PR-AUC 0.082 ± 0.002 versus random 0.206 ± 0.006 ($+0.124 \pm 0.007$). Notably the base-rate effect runs *against* this inflation (the temporal test window has the *higher* prevalence), so the $+0.124$ is distribution access net of an adverse base-rate term. Across four datasets the inflation thus tracks the drift magnitude: stable (DGraph) $+0.001$; moderate drift (AMLworld) $+0.124$; regime change (Elliptic family) $+0.41$ to $+0.55$. Three points across different domains and model classes are a consistent pattern rather than a fitted law, but the dose-response direction is exactly what the distribution-access mechanism predicts.

4.6 The unsolved post-shift regime and negative method results

Under the temporal protocol, a rolling per-window backtest of EvolveGCN-O [24] collapses at the dark-market shutdown (steps 44–46: PR-AUC ~ 0.01 ; Figure 3) and only partially recovers. A learning-rate $\times$ gradient-clipping sweep (six configurations, seed 42) improves the aggregate test PR-AUC from 0.100 to at best 0.200 ($1r\ 5 \times 10^{-3}$, no clipping), still far below every temporal-protocol baseline; the conclusion is unchanged.

Is the post-shift regime merely unexplored, or genuinely hard? We tested four drift-robust variants; none improves on the baselines. A USAD-style two-decoder adversarial graph autoencoder [17] (naive, rolling-normal, and domain-adversarial variants) scores PR-AUC 0.0367 ± 0.0001 over three seeds, a stable and decisive negative. A supervised domain-adversarial GraphSAGE [25] scores

0.462 $\pm$ 0.090 over three seeds, statistically indistinguishable from the plain baseline (0.510 $\pm$ 0.039); the time-invariance constraint yields no robustness gain. We note a second self-correction here: at a single seed (42, PR-AUC 0.364) DANN appeared clearly harmful; the multi-seed run shows that stronger conclusion was seed noise. Enforcing time-invariance does not solve this regime change; whether it actively hurts is within noise.

5 Discussion and recommendations

(1) On temporally shifting fraud graphs, report illicit-class metrics under a strict temporal split; random-split numbers can overstate PR-AUC by up to +0.55. (2) The post-shift window is the real benchmark; aggregate scores hide the collapse. (3) Graph structure is not a free win on Elliptic: features already encode neighborhood aggregates, and under honest evaluation trees lead. (4) Method claims should be accompanied by decomposition-style controls; our own double-leak hypothesis did not survive its ablation.

6 Limitations

(1) The tuning sweeps and the EvolveGCN sweep are single runs at seed 42; the headline inflation, decomposition, DGraph control, and negative-method results are multi-seed (3-5 seeds). (2) The DGraph-Fin control operates near the PR-AUC floor for the models we trained (PR-AUC 0.036–0.040 against a 1.3% base rate), ROC-AUC does inflate there (+0.025 $\pm$ 0.006 over seeds; up to +0.055 at width 128), and GADBench reports far stronger GNNs on this dataset; the control is therefore supporting evidence for the shift-driven mechanism, not proof. (3) Our temporal-protocol GNN F1 sits below some honest-protocol literature values ($\sim$ 0.69 for GraphSAGE) even after tuning; the trees-over-GNNs magnitude we measure is therefore larger than the field’s, though the ordering is consistent. (4) The trees-over-GNNs ordering is established here on Elliptic-family data only. (5) The GAT equivalence bound on message-passing leakage is loose ($\pm$ 0.05). (6) DGraph-Fin’s temporal stability is measured coarsely (labeled fraud prevalence 1.2% in the train window versus 1.4% in the test window under our first-seen split), not characterized in depth. (7) Per concurrent work [15], Elliptic’s features embed construction-level look-ahead (address aggregates computed after the test period) and entity nonindependence; our “honest” protocol is honest at the *split* level only. Both arms share the same features, so exposure is symmetric; the induced effect on each arm need not be, so construction-level leakage primarily qualifies our absolute temporal-arm scores (upper bounds on deployable performance) and could second-order perturb the arm differences on which the decomposition rests.

7 Conclusion

The near-perfect reported performance on Elliptic-family benchmarks is largely an evaluation artifact whose dominant mechanism is distribution access, not graph-specific leakage. We provide the decomposition, the controls, a reproducible harness together with negative results reported in full, and we re-anchor the state of the art under the temporal protocol.

Reproducibility statement

Every number in this paper is produced by one seeded CLI command. The anonymized supplementary archive contains the full source code, the per-seed values for every cell, and the statistical summary (docs/results/stats_summary.json). The public repository, which additionally carries a continuous-integration pipeline (lint, typecheck, tests, Docker smoke) on every commit, is linked in the camera-ready version. Elliptic downloads automatically via PyTorch Geometric; Elliptic++ and DGraph-Fin are obtained from their authors (DGraph-Fin is gated and requires registration); AMLworld is obtained from Kaggle.

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A Per-seed values for the headline inflation

Table 5 lists the per-seed PR-AUC values behind Table 2; the released results files contain the same per-seed detail for every other cell in the paper.

Table 5: Per-seed illicit-class PR-AUC (seeds 42–46).

Model	Arm	s42	s43	s44	s45	s46
GCN	temporal	0.294	0.324	0.206	0.201	0.302
	random	0.800	0.820	0.824	0.822	0.823
GraphSAGE	temporal	0.488	0.542	0.455	0.546	0.522
	random	0.925	0.912	0.930	0.913	0.922
GAT	temporal	0.332	0.251	0.285	0.236	0.384
	random	0.811	0.783	0.746	0.730	0.806
GraphTransformer	temporal	0.487	0.436	0.498	0.475	0.540
	random	0.897	0.899	0.912	0.909	0.896
XGBoost	temporal	0.790	0.790	0.793	0.792	0.790
	random	0.987	0.984	0.986	0.986	0.984

B GenAI usage statement

Code, experiments, and manuscript drafting were carried out with substantial assistance from an LLM-based coding agent (Claude, Anthropic), with the agent workflow documented in the repository. The research questions, methodological decisions, and final claims were set and verified by the human author, who takes full responsibility for the content.